Unit 4

Magnetic Effect of Electric Current

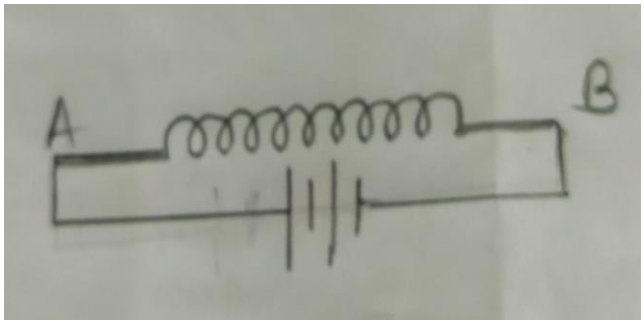
Answer the following questions
Each question carries 1 score.

1.Find the odd one and write down the reason
(Paper diaphragm, voice coil, split rings,
field magnet)

2.A current carrying solenoid is stretched to increase the distance between the coils, what change will occur in its magnetic field?
(Increases/Decreases)

3.When electricity flows through a straight conductor the rule used to detect the direction of the magnetic field around it is?

4.Observe the figure.



a. What is the magnetic polarity of end A?

5. Assertion:The magnitude of the magnetic field at a point on the axis of the current carrying solenoid is inversely proportional to the current flowing through the solenoid.
Reason: the magnitude of the magnetic field at a point on the axis of the current carrying solenoid is directly proportional to the number of turns per unit length of a solenoid.

a) Both Assertion and reason are true and the reason is the correct explanation of assertion.
b) Both assertion and reason are true but reason is not the correct explanation of the assertion.
c) Assertion is true but reason is false.
d) Assertion is false but reason is true.
e) Both assertion and reason are false.

6. In Fleming's left hand rule the thumb represents the direction of……

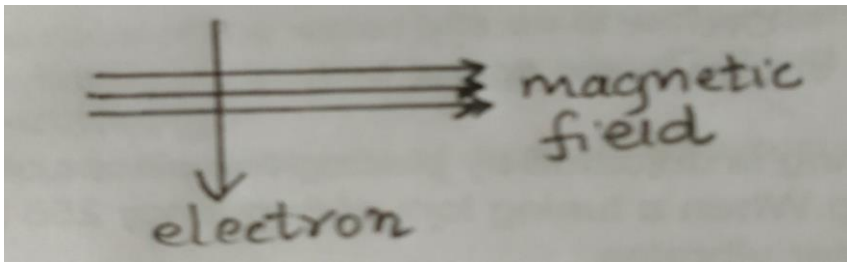
Answer the following questions:
Each question carries 2 scores

7. Write down whether the given statements are right or wrong, correct the incorrect statement.

a) If the direction of the current is clockwise at one end of the solenoid then that end will be the north pole.
b) If the direction of the current is anti clockwise at one end of the solenoid then that end will be the South Pole.
c) In a moving coil loudspeaker the electrical energy is converted to sound energy.

8. In a split ring commutator of a DC motor semicircular rings are used. What is the need for this?

9 An electron enters a magnetic field at right angles to it as shown in the figure,



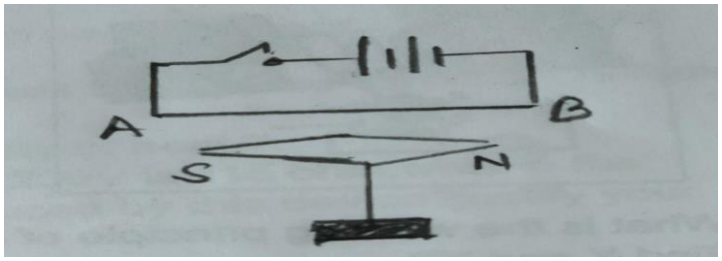
a)the direction of the force acting on the electron will be.

a) to the right
b) to the left
c) out from the page
d) into the page

b) State the law which is helped to find the direction of force experienced by the electrons.

10. A current carrying solenoid is stretched to increase the distance between the coils . What change will occur in its magnetic field.? Describe.

11.Observe the figure.



a) In which direction does the north pole of the magnetic needle deflect when the switch in the circuit isl turned ON?

(Clockwise, anticlockwise)

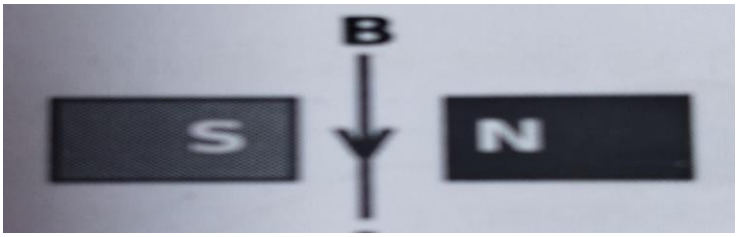
b) Suggest two ways to make the direction of the north pole of the magnetic needle in the opposite direction.

12.a) What happens when we place a magnetic compass nearer to the current carrying conductor.

b) Give reason

13.Define a solenoid. Write any one similarity between the magnetic field of a solenoid and that of a current carrying conductor.

14. A current carrying straight conductor is placed between two poles of a magnet as shown.



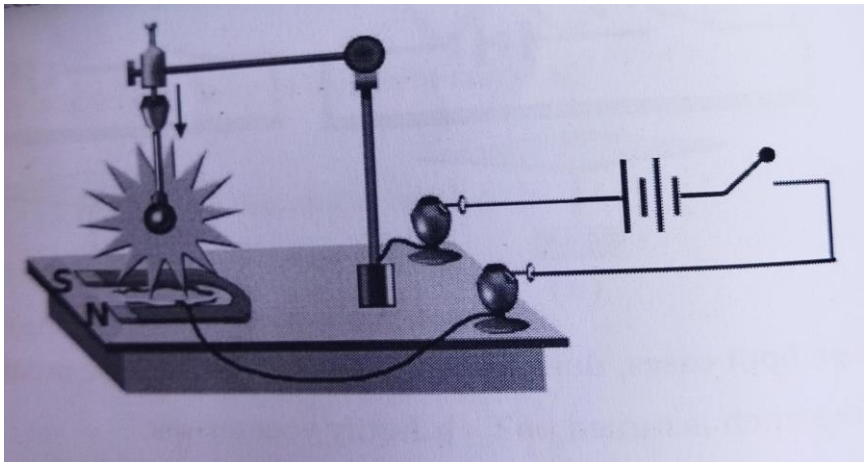
a) The direction of magnetic force on the conductor will be…..
(Outward from the plane of paper / into the plane of paper)

b) State the rule used to find the answer.

Answer the following questions

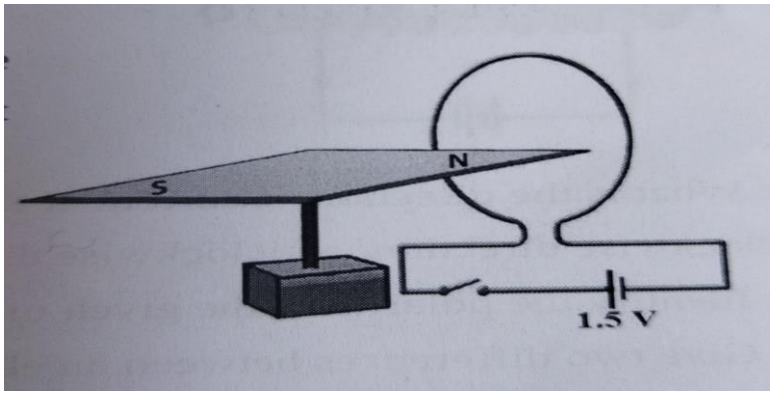
Each question carries 3 scores

15.A wooden block contains mercury between the north and south poles. A freely rotating toothed wheel is in contact with the mercury. When an electric current is passed through the wheel,



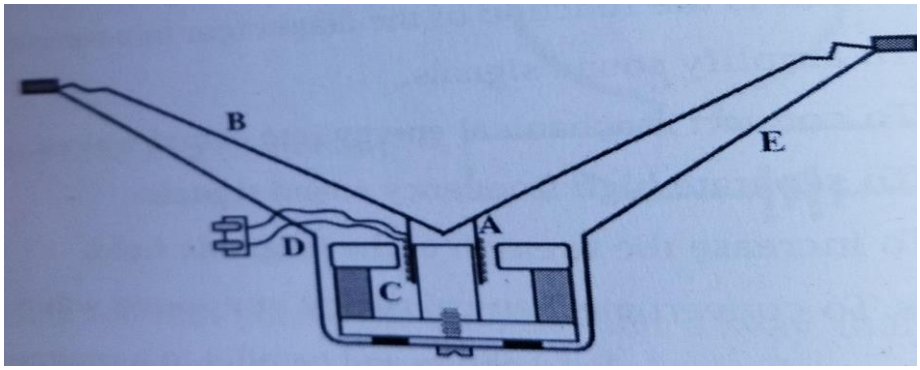
a) In which direction the wheel is rotating?
(Clockwise direction/anticlockwise direction)
b) Justify your answer.

16.A magnetic compass is placed near a single coil. When the switch is turned on, the magnetic needle deflects. What happens to the deflection in each of the following cases?



a) A coil with more turns is used.
b) The terminals of the battery are interchanged.
c) The applied voltage is increased.

17.Observe the figure



a) Identify the device shown in the diagram.
b) What is its working principle?
c) What is the energy conversion taking place in this device?

18. Moving coil loudspeaker convert electrical energy in to sound energy

a) Where is the voice coil situated?

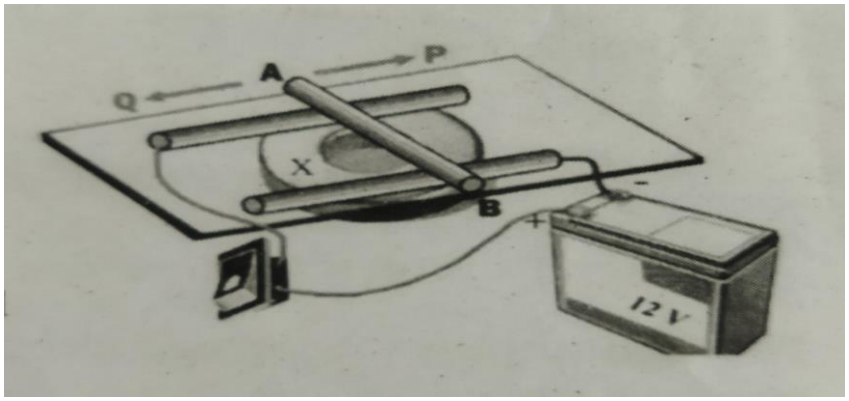
b) From where does the electric current reach the coil?

c) To which part is the diaphragm connected?

Answer the following questions

Each question carries 4 scores

19. In the figure a conductor AB is arranged across two parallel copper wires placed on an acrylic sheet which is kept above a ring magnet. A 12 V battery is connected to the parallel conductors through a bell switch.

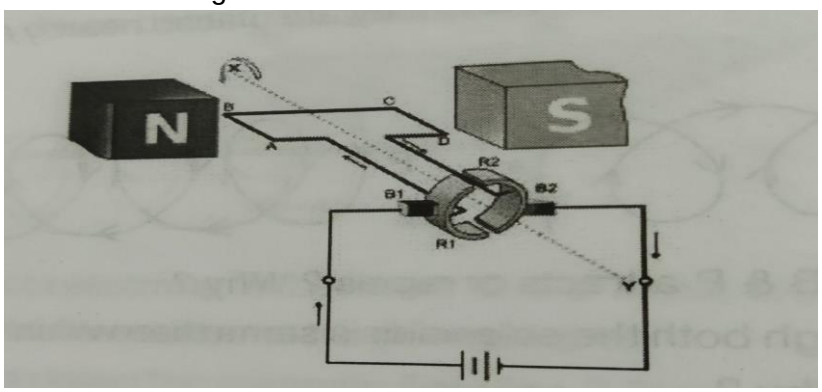


a) Which magnetic pole should be at the position of X for the conductor AB to move towards P?

b) Which law helps in arriving at the answer? state the law .

c) What change must be made to retain the initial direction of motion of the conductor when the polarity of the magnet is reversed?

20. Observe the figure



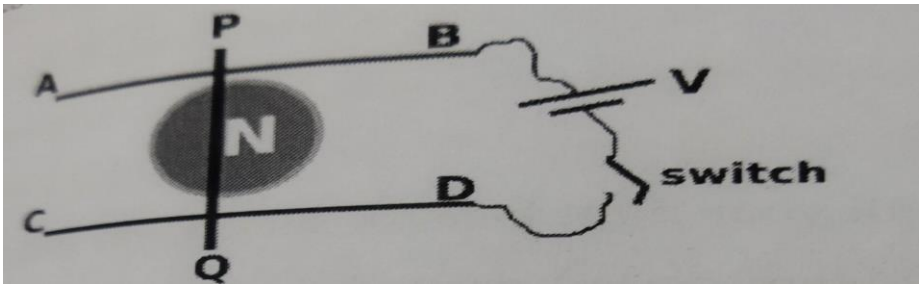
a) Identify the device

b) What is the direction of force acting on the arm CD in the figure? (upwards/downwards)

c) Does the force acting on the arms AB and CD are in the same direction? Why?

d) What is the function of split ring commutator in this device?

21.An experimental setup demonstrated in class is shown.



AB and CD are two non insulated copper rods fixed on a cardboard. PQ is another non insulated copper rod that can move along them. A disc magnet is placed below the rod PQ with its north pole facing upward.

a) When the switch is turned on PQ will move towards (PA/PB)
b) Name two devices that work on the basis of this phenomenon.
c) Explain this phenomenon.

Answer key

1.Split rings

Split rings are part of the electric motor.The paper diaphragm, voice coil and the field magnet arevthe parts of a moving coil loud speaker.

2.Decreases.

3. Right Hand Thumb Rule

4.South ploe

5.d. Assertion is false but reason is true.

6.Direction of current.

7. a) Incorrect statement. if direction of current is clockwise at one end of the solenoid then that end will be the South Pole.

b) Incorrect statement. If direction of current is anticlockwise at one end of the solenoid then that End will be the north pole.

c) Correct.

8. In a motor the split ring a commutator is rotating along with the armature. The position of the half rings in the split ring commutator changes continuously. So the direction of current through the coil changes after every half rotation thus contains rotation is possible in a motor.

9.a) Into the page

b) Fleming's left hand rule.

10. The strength of magnetic field decreases. Ask the number of magnetic field lines per unit area decreases the Magnetic Strength decreases.

11.a) anti clockwise.

b) reverse the current

Arrange the conducting wire AB below the magnetic needle.

12.a) Needle of the magnetic compass deflects.

b) it is due to the interaction between the magnetic field of the magnetic compass and that of the magnetic field produced by the current flow .

13. An insulated conductor wound in a spiral shape is called a solenoid.

Both the magnetic field of a solenoid and that of a current-carrying conductor are produced due to the flow of electric current.

14.a) into the plane of the paper

b) Fleming's left hand rule.

Hold the thumb,first finger and second finger of your left hand perpendicular to each other. If the first finger points in the direction of the magnetic field and the second finger in the direction of the electric current then the thumb will indicate the direction of the force experienced by the conductor.

15.a) Anticlockwise direction.

b) When the switch is turned on the current flows downward through the wheel to the Mercury. According to the motor principle the toothed wheel experience a force directed such that it rotate in the anti clockwise direction as per Fleming's left hand rule.

16.a)deflects more

b)deflection reverses

c)deflects more.

17 a) moving coil loudspeaker.

b) motor principle

c) converts electrical energy into sound energy.

18.a) in between the poles of the magnet.

b) from amplifier

c) voice coil

19.a) South

b) Fleming’s left hand rule

Statement:

c) Change the direction of electric current

20.a) Motor

b) upwards

c) No. In opposite direction. The direction of current through AB and C D are opposite.

d) The split rings help to change the direction of current through the coil after every half rotation.

21.a) PA

b)Electric motor and moving coil loudspeaker.

c) A current carrying conductor which is free to move, placed in a magnetic field, exhibits at tendency to deflect. This is Motor principle.